

JASON TIANCHI FAN

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EDUCATION

Master of City Planning

University of Pennsylvania, Philadelphia PA

Expected May 2026

- Marketing lead of the Student Planning Association and the Wharton-Weitzman Future of Cities Conference; Student Admissions Ambassador; Teaching Assistant for CPLN 7900 MUSA/Smart City Practicum

Bachelor of Arts with distinction, Major in Political Science

Simon Fraser University, Burnaby BC

Sept 2019 – May 2024

WORK EXPERIENCE

Geospatial Researcher

Penn Institute of Urban Research, Philadelphia PA

Jan 2026 – Present

- Designed and executed geospatial data workflows in ArcGIS and Python to quantify Penn's institutional footprint, integrating socio-economic datasets to model spatial influence across regional geographies.
- Built data integration pipelines combining parcel-level, census, and administrative datasets to support multi-variable socio-economic assessments informing university-wide development frameworks.
- Translated modeling outputs into visualizations and reports for university stakeholders, communicating spatial patterns and analytical findings to both technical and non-technical audiences.

Design Fellow

PennPraxis, Philadelphia PA

May 2025 – Oct 2025

- Built and maintained scalable data pipelines in R to ingest, clean, and integrate large-scale microdata for 245,000+ individuals from multiple sources (US Census PUMS, Data Axle), ensuring data quality and analytical readiness for statewide economic modeling.
- Applied descriptive and diagnostic statistical techniques to identify regional trends in self-employment, wage disparities, and industry growth, translating modeling outputs into actionable insights for statewide economic development policy.
- Operationalized analytical findings into a published, publicly distributed report, designing data visualizations and media assets in Adobe InDesign and Illustrator to communicate results to both technical and non-technical stakeholders.

Knowledge Mobilization Lead

Department of Gerontology, Simon Fraser University, Burnaby BC

Jan 2024 – Dec 2024

- Collected and processed spatial audit data across 20km+ of pedestrian infrastructure using the SWAN tool, performing quality assurance on field data to identify critical accessibility gaps and support age-friendly urban design recommendations.
- Utilized NVivo and photo-voice methodologies to analyze the spatial and social lived experiences of older adults, translating qualitative insights into actionable "aging-in-place" strategies.
- Orchestrated civic education and knowledge mobilization by designing a board game, professional video content, and conference posters to communicate complex research to public audiences.

SELECT PROJECTS

Pennsylvania Creative Workforce Analysis | Publication

Oct 2025

- Designed a reproducible R-based pipeline to integrate and transform multi-year PUMS and commercial microdata, enabling cross-state benchmarking of creative sector employment trends; results published by the Pennsylvania Council on the Arts.

PhillyStat360: Vacant Property Risk Model | Practicum

In Progress

- Built a machine learning pipeline in R to predict parcel-level vacancy risk across ~352K Philadelphia residential properties, integrating eight administrative datasets from OPA and L&I into a unified feature matrix with engineered life-history, transfer, and structural distress signals.

Proactive Imminently Dangerous Building Detection | Course Project

In Progress

- Designing a semantic segmentation pipeline using the Clay Vision Transformer foundation model fine-tuned in PyTorch on Philadelphia's 0.25ft aerial imagery to generate parcel-level structural distress scores, integrating temporal change detection across 2024–2025 imagery to flag deteriorating structures for L&I Emergency Services prioritization.

SKILLS

Languages & Tools: Python, R, PostgreSQL | **Platforms:** ArcGIS, QGIS | **Libraries:** Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, Matplotlib, OSMnx | **Visualization:** Illustrator, InDesign | **Qualitative:** NVivo | **Languages:** English, Mandarin